ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** neuron	Neuron.
2	**(c)** insulin	Insulin.
3	**(d)** phototropism	Phototropism.
4	**(c)** axon	Axon.
5	**(b)** thyroid	Thyroid.
6	**(c)** water and gravity	Hydrotropism and geotropism.
7	**(c)** sensory neurons	Sensory neurons.
8	**(a)** adrenal glands	Adrenal glands above kidneys.
9	**(c)** auxin	Auxin.
10	**(d)** motor neurons	Motor neurons.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Nerve cell (1); Dendrite, cell body, axon (1).	1+1
12	Forebrain (1); Midbrain and hindbrain (1).	1+1
13	Directional growth movement in response to a stimulus (1); Phototropism and geotropism (1).	1+1
14	A change in the environment that causes a response (1); Heat, light, touch (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Voluntary: brain-controlled (1); Reflex: spinal cord (1); Reflex is faster (1).	1+1+1
16	Insulin: pancreas (1); Thyroxine: thyroid (1); Adrenaline: adrenal (1).	1+1+1
17	Prepares the body for emergency (1); Increases heart rate (1); Fight or flight (1).	1+1+1
	OR: Receptor (1); Sensory neuron, spinal cord, motor neuron (1); Effector (1).	1+1+1
18	Goitre (1); Diabetes (1); Dwarfism (1).	1+1+1
	OR: Differential growth (1); Touch stimulates (1); Helps the plant climb (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dendrite receives information (1); Cell body has nucleus (1); Axon carries impulse (1); Nerve endings pass impulse (1); Synapse between neurons (1).	1+1+1+1+1
	OR: Diabetes: insulin (1); Goitre: iodine/thyroxine (1); Dwarfism: growth hormone (1); Gigantism: excess (1); Treatments (1).	1+1+1+1+1
20	Example: touching hot object (1); Receptor to sensory neuron (1); Spinal cord (1); Motor neuron to muscle (1); Hand withdrawn quickly (1).	1+1+1+1+1
	OR: Fast vs slow (1); Electrical vs chemical (1); Neurons vs blood (1); Short vs long effect (1); Examples (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a reflex action is controlled by the brain; the reflex arc passes through the spinal cord (the brain is informed later).
- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
- Confusing gland and hormone pairs: adrenaline is from the adrenal gland (not the pituitary), insulin from the pancreas, thyroxine from the thyroid.
- Mixing up the functions of brain parts: cerebellum maintains posture and balance; medulla controls involuntary actions such as blood pressure and vomiting.
- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.

- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** pituitary	Pituitary.
2	**(d)** cytokinin	Cytokinin.
3	**(c)** ear	Ear.
4	**(b)** iodine	Iodine.
5	**(d)** abscisic acid	ABA.
6	**(a)** tongue	Tongue.
7	**(d)** insulin	Insulin.
8	**(d)** thigmotropism	Touch.
9	**(c)** brain	Brain.
10	**(b)** testosterone	Testes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gap between two neurons (1); Impulse passes by chemicals (1).	1+1
12	Auxin: elongation (1); Cytokinin: cell division (1).	1+1
13	Cells that detect stimulus (1); Eye and ear (1).	1+1
14	Chemical messenger secreted by glands (1); Insulin, thyroxine (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Eye: light (1); Ear: sound (1); Nose: smell (1).	1+1+1
16	Phototropism: light (1); Geotropism: gravity (1); Hydrotropism: water (1).	1+1+1
17	Dendrite tip (1); Cell body (1); Axon then nerve ending (1).	1+1+1
	OR: Thinking and memory (1); Controls voluntary actions (1); Hunger centre (1).	1+1+1
18	Growth of stem (1); Cell division (1); Wilting of leaves (1).	1+1+1
	OR: Secretes insulin (1); Regulates blood sugar (1); Also secretes digestive juice (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cerebrum: thinking (1); Cerebellum: balance (1); Medulla: involuntary actions (1); Pons: relay (1); Midbrain: eye reflexes (1).	1+1+1+1+1
	OR: Brain and spinal cord (1); Cranium (1); Vertebral column (1); Meninges (1); CSF (1).	1+1+1+1+1
20	Phototropism: shoot to light (1); Geotropism: root down (1); Hydrotropism: root to water (1); Chemotropism: pollen tube (1); Thigmotropism: tendril (1).	1+1+1+1+1
	OR: Sugar rises: insulin released (1); Sugar falls: insulin falls (1); Pancreas senses it (1); Diabetes if insulin is low (1); Treatment by insulin injection (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a reflex action is controlled by the brain; the reflex arc passes through the spinal cord (the brain is informed later).
- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
- Confusing gland and hormone pairs: adrenaline is from the adrenal gland (not the pituitary), insulin from the pancreas, thyroxine from the thyroid.
- Mixing up the functions of brain parts: cerebellum maintains posture and balance; medulla controls involuntary actions such as blood pressure and vomiting.
- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.

- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Mimosa pudica	Touch-me-not.
2	**(a)** cerebrum	Cerebrum.
3	**(b)** oestrogen	Ovaries.
4	**(b)** shoot and root tips	Meristems.
5	**(b)** reflex action	Reflex.
6	**(b)** blood	Blood.
7	**(a)** tropism	Tropism.
8	**(d)** writing	Writing.
9	**(b)** hormones	Hormones.
10	**(a)** chemicals	Chemotropism.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Maintains balance (1); Controls posture (1).	1+1
12	Sudden involuntary response to a stimulus (1); Withdrawing hand from a hot object (1).	1+1
13	Ductless glands (1); Thyroid, pancreas (1).	1+1
14	Controls involuntary actions (1); Like blood pressure, salivation, vomiting (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Skull (1); Meninges (1); Cerebrospinal fluid (1).	1+1+1
16	Iodine makes thyroxine (1); Prevents goitre (1); Needed for growth (1).	1+1+1
17	Collects on the shaded side (1); Cells elongate more (1); Shoot bends to light (1).	1+1+1
	OR: Speed: fast vs slow (1); Medium: neurons vs blood (1); Duration: short vs long (1).	1+1+1
18	Leaves fold on touch (1); Water leaves cells (1); Not due to growth (1).	1+1+1
	OR: Carries messages to and from the brain (1); Controls reflexes (1); Protected by the vertebral column (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Insulin (1); Thyroxine (1); Adrenaline (1); Growth hormone (1); Testosterone/oestrogen (1).	1+1+1+1+1
	OR: Sensitive plant: leaves fold (1); Tendril: coils round support (1); Sensitive plant: turgor change, fast (1); Tendril: differential growth, slow (1); Both are responses to touch (1).	1+1+1+1+1
20	Auxin (1); Gibberellin (1); Cytokinin (1); Abscisic acid (1); Ethylene note (1).	1+1+1+1+1
	OR: Shoot tip senses light (1); Auxin goes to shaded side (1); Cells elongate (1); Shoot bends (1); Positive phototropism (1).	1+1+1+1+1

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- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.
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MASTERMIND